

Summary of the Article

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A Survey of Machine Learning Techniques for Artificial Intelligence

The article "*A Survey of Machine Learning Techniques for Artificial Intelligence*" provides a comprehensive overview of different machine learning techniques used in the development of artificial intelligence systems. The paper explains that machine learning is a core part of AI that enables systems to learn from data, identify patterns, and make decisions without being explicitly programmed.

The study categorizes machine learning into four main types: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. Each of these techniques is explained in detail along with their working principles and applications.

Supervised learning is used where labeled data is available, and it includes algorithms such as linear regression, decision trees, neural networks, and support vector machines. These methods are widely used in classification and prediction tasks like image recognition and medical diagnosis.

Unsupervised learning works with unlabeled data and focuses on finding hidden patterns. Techniques such as K-means clustering, hierarchical clustering, and dimensionality reduction methods like PCA and t-SNE are discussed in the paper.

Semi-supervised learning combines both labeled and unlabeled data to improve learning accuracy. Methods like self-training, co-training, label propagation, and generative models are highlighted as important approaches when labeled data is limited.

Reinforcement learning is another important area where an agent learns through interaction with an environment using rewards and penalties. Techniques such as Q-learning, Deep Q-Networks (DQN), PPO, and DDPG are discussed as powerful methods for decision-making tasks.

In conclusion, the paper emphasizes that machine learning techniques are essential for the growth of artificial intelligence. These methods allow AI systems to become more intelligent, adaptive, and capable of solving complex real-world problems in areas such as healthcare, robotics, cybersecurity, and automation.

Reference:

Rathod, Y. (2024). *A Survey of Machine Learning Techniques for Artificial Intelligence*. International Journal of Computer Techniques. Available at: ResearchGate.

Available at:

https://www.researchgate.net/publication/382264507_A_Survey_of_Machine_Learning_Techniques_for_Artificial_Intelligence